Unit 4

Introduction to Demand and Supply

Introduction

You may have questions such as next: What does demand mean? What does the law supply say? What does market equilibrium mean? The purpose of this unit is to explain what demand and supply are and show how they determine equilibrium price and quantity. We will also show how the concepts of demand and supply reveal consumers and producers sensitivity to price change. This unit deals with the concept of demand, the concept of supply and market equilibrium.

Unit objectives

At the end of this unit, students will be able to:

- Define the concepts of demand and supply.
- Explain the laws governing demand and supply.
- Describe the equilibrium price and quantity

Start-up Activities

1. How do you describe the term demand?
2. How do you define the term 'supply' in economics?
3. Can you define the concept of 'equilibrium'?

Key Concepts

Ceteris paribus, demand, demand schedule, demand curve, demand function, law of demand, supply, supply schedule, supply curve, law of supply, equilibrium, market demand, market supply, market equilibrium, equilibrium price, and equilibrium quantity.

4.1 Concept of Demand

At the end of this section, students will be able to:

- Define the concept of demand
- Analyse the law of demand

Start-up Activity

How do you describe the term demand?

Demand

The terms demand, desire and want are frequently used synonymously to express what an individual needs and would like to acquire. Demand states that the consumer must be willing and able to buy a commodity which he or she desires at a given price during a given period of time.

Accordingly, demand is distinct from a mere desire to acquire something. Human wants are unlimited and desires are numerous. However, only a desire backed up by the ability to pay the price for the commodity and the willingness to purchase it, is called a demand. We can say demand refers to an effective desire. A desire becomes an effective demand only when it is backed by the following three features:

- ability to pay for the good desired,
- willingness to pay the price of the good desired, and
- availability of the good itself

Furthermore, demand for a good is constantly stated relative to a specific price and certain time. For instance, an individual may be interested to buy a certain jeans at a price of Birr 500, but he or she could not absolutely demand it if its price is Birr 900. Likewise, an individual may be willing to buy a room heater at a price of Birr 1000 during a cold season, but he or she may not be interested in buying it at this price during a hot season.

Based on the aforementioned considerations, we can define demand as follows:

> Demand for a commodity is the amount of it that a consumer is willing to buy at various given prices and a given moment of time.

Quantity Demanded

Quantity demanded is the amount of commodity a consumer must be willing and able to buy which he/she desires at a given price during a given period of time.

Law of Demand

Law of demand states that, ceteris paribus, price of a commodity and its quantity demanded are inversely related. Ceteris paribus means other thing remain the same. In other words, the higher the price, the lower the quantity demanded. The law of demand is a fundamental principle of economics that states that at a higher price consumers will demand a lower quantity of a good.

Demand Schedule

A demand schedule is a tabular description, which presents various quantities of a commodity that would be demanded at different prices. Demand schedule refers to a tabular representation of the relationship between price and quantity demanded. It demonstrates the quantity of a product demanded by an individual or a group of individuals at specified price and time.

Table 4.1: Individual household demand schedule for Oranges per week

Price (Birr Per kg)	Quantity Demanded/week
5	5
4	7
3	9
2	11
1	13

The above demand schedule shows the different quantities of oranges demanded by an individual household at different prices. At Birr 5 per kg, the consumer demands 5kg of oranges. However, an individual household's demand becomes 13 kg of Oranges at Birr 1 per kg.

Demand Curve

A demand curve is a graphical representation of the relationship between different quantities of a commodity demanded by an individual at different prices per time period.

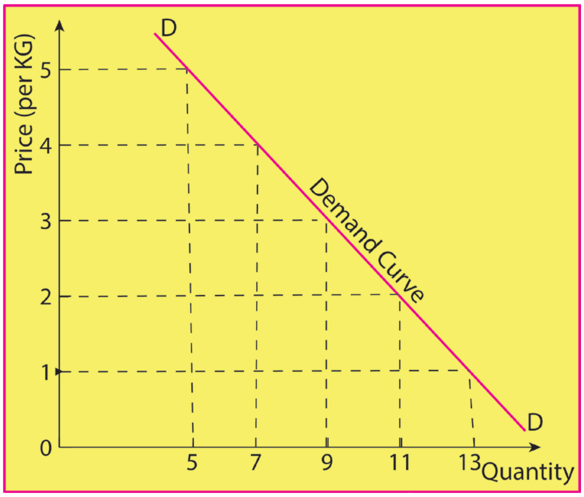

Figure 4.1: Individual household demand curve

Demand Function

It is a mathematical representation of the relationship between price and quantity demanded, ceteris paribus. A typical demand function is given by:

$Q_d = f(P)$

where, Q_d is the quantity demanded and P is the commodity's price,

Market demand

Market demand describes the demand for a given product and who wants to purchase it. This is determined by the willingness of consumers to spend a certain price on a particular good or service. As market demand increases, price also increases. When it decreases, price will decrease as well. Market demand is derived by a simple horizontal summation of the quantity demanded for a commodity by all buyers at each price.

The market demand curve describes the relationship between various quantities of a commodity that consumers are willing to buy at different prices. Market demand curve is the horizontal summation of individual demand curves at the market price.

Activity 4.1

1. Define demand and state the law of demand.
2. Explain why the demand curve always slopes downwards from left to right.

4.2 Concept of Supply

At the end of this section, students will be able to:

- Define the concept of supply.
- Analyse the law of supply

Start-up Activity

How do you define the term 'supply' in economics?

Supply

In ordinary language the term supply is often misused and confused with the term 'stock'. Stock is the total volume of a commodity produced during a period less the quantity already sold out. Conversely, supply means the quantity which is actually brought to the market. Most of the time, producers do not offer all of their stock for sale in the market. For example, at the time of harvest, a large part of agricultural product is kept back in cold storage and is taken out during the off-season for sale at better prices. Likewise, a part of industrial product is usually kept back in stock and is offered for sale in the market at the time when it can bring higher prices. Thus, we can say that stock is potential supply and supply may be less than or equal to the stock of a commodity. However, in economics, the term supply has a specific meaning and it is described as:

> Supply of a commodity refers to the various quantities of it which producers are willing and able to offer for sale at a particular time at various corresponding prices.

Quantity Supplied

Quantity Supplied indicates various quantities of a commodity that sellers (producers) are willing and able to provide at different prices in a given period of time. In economics, quantity supplied describes the number of goods or services that suppliers will produce and sell at a given market price. The quantity supplied differs from the actual amount of supply as price changes influence how much supply producers actually put on the market.

Note that supply shows the relationship between the quantity supplied and price of a commodity, while quantity supplied refers to a specific amount of the commodity, which a producer is willing to sell at a specific price.

Law of Supply

Like the law of demand, the law of supply demonstrates the quantities sold at a specific price. But unlike the law of demand, the supply relationship shows an upward slope. This means that the higher the price, the higher the quantity supplied. The law of supply states that, ceteris paribus, there is a direct or positive relationship between price and the quantity supplied of a commodity.

Supply Schedule

A supply schedule is a tabular presentation of different quantities of a commodity offered for sale at different prices per time period. Supply schedule is a chart that shows how much product a supplier will have to produce to meet consumer demand at a specified price based on the supply curve.

Table 4.2: Individual seller's supply schedule for oranges

Price (Birr Per kg)	Quantity Supplied (kg/week
10	60
15	70
20	80
25	90
30	100

The above demand schedule shows the different quantities of Oranges demanded by an individual household at different prices. At Birr 5per kg consumer demands 5kg of Oranges. However, an individual household's demand becomes 13 kg of Oranges at Birr 1 per kg.

Supply curve

A supply curve shows the information of a seller's supply schedule graphically rather than in a tabular form.

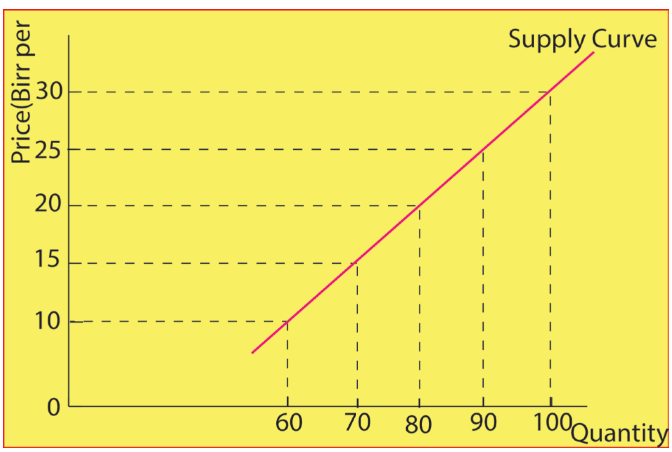

Figure 4.2 Individual seller's supply curve for oranges

Supply Function

A supply function is a mathematical representation of the relationship between price and quantity supplied of a commodity, all other things remaining the same. A typical supply function is given by:

$Q_S = f(P)$

where, Q_S is quantity supplied and P is the price of the commodity.

Market supply

Market supply is the total amount of an item producers are willing and able to sell at different prices, over a given period of time. Market supply is derived by horizontally adding the quantity supplied of the commodity by all sellers at each price.

The market supply curve indicates the relationship between various quantities of a commodity that sellers are willing to offer for sale at different prices. Market supply curve is the horizontal summation of individual supply curves at the market price.

✎ **Activity 4.2**

1. Define supply and state the law of supply.
2. Why does the supply curve slope upward to the right?

4.3 Market Equilibrium

At the end of this section, students will be able to:

- Define market equilibrium

Start-up Activity

Can you define the concept of 'equilibrium'?

In the preceding sections, we have discussed both the consumers' demand for goods in a market and the firms' supply of goods in a market independently. Demand and supply curves tell us how much consumers demand and how much producers supply at different prices, respectively. However, they do not tell us the actual price of a good in general. We now come to the important question: How do the forces of demand and supply work together in order to determine market prices? We explain how forces of demand and supply help in attaining 'equilibrium', and how the 'equilibrium price' is determined.

The concept of equilibrium

The term "equilibrium" refers to a state of balance. In the physical world, when two opposing forces acting on an object are balanced so that the object is held stationary, the object is said to be in equilibrium. In simple terms, when the object under the action of forces working in opposite directions has no tendency to move in any direction, the object is in equilibrium. Similarly, an economic system is said to be in equilibrium when its important variables show no change, and when no forces are acting on them to produce a change in their values. For instance, after attaining equilibrium, a consumer has no intention of re-allocating his or her money or income. Likewise, a firm is said to be in equilibrium when it has no tendency to change its level of output by either increasing or decreasing it. Thus, there is a tendency to move towards the equilibrium price. This tendency is known as the "market mechanism," and the resulting balance between supply and demand is called "market equilibrium."

In real world economic activities, equilibrium may never be actually realised. The central feature of equilibrium analysis in economics is the concept that economies tend towards equilibrium when no external forces are acting on them.

Market equilibrium

In the context of price determination, equilibrium refers to a situation in which the quantity demanded of a commodity equals the quantity supplied of the commodity. In brief, it refers to the balance between opposite forces of demand and supply known as market equilibrium.

Equilibrium price

The price at which the quantity demanded of a commodity equals the quantity supplied is called 'equilibrium price'. The forces of demand and supply determine the price of a commodity in a given market. At an equilibrium price, the quantity demanded and the quantity supplied are equal.

Equilibrium quantity

Equilibrium quantity refers to the quantity of a good supplied in the marketplace when the quantity supplied by sellers exactly matches the quantity demanded by buyers. In other words, the equilibrium quantity is the amount of a commodity that is bought and sold at an equilibrium price.

Graphically, we depict market equilibrium as shown in the Figure below.

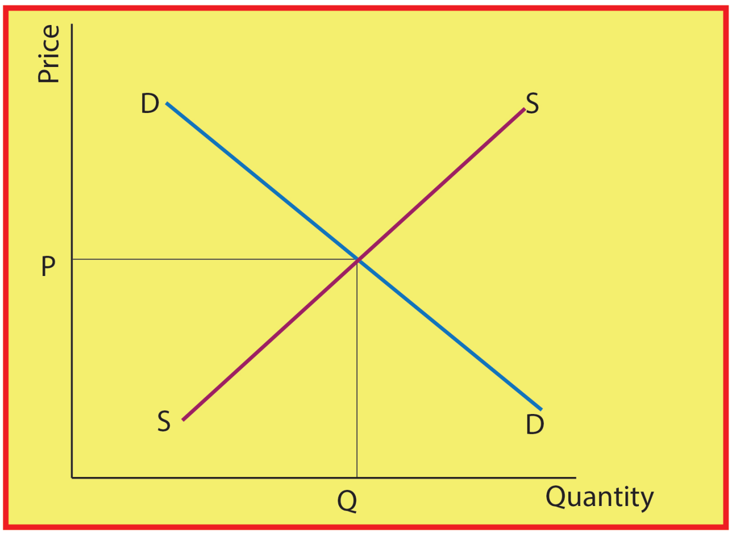

Figure 4.3 Market equilibrium

When the demand curve and the supply curve intersect, we have a point where the quantity that consumers are willing to purchase matches the quantity that suppliers are willing to supply at a given price. This point is known as the market equilibrium. From the market equilibrium we can derive market price and market quantity.

Note: This market equilibrium is not fixed; it is likely to change over time due to changes in the patterns of demand and supply.

Activity 4.3

1. What is the meaning of market equilibrium?
2. Explain the difference between equilibrium price and equilibrium quantity.

Unit Summary

Demand for a commodity is the amount that will be purchased at a particular price during a particular period of time. The law of demand states that, other things remaining constant, the quantity demanded of a commodity increases when its price falls and decreases when it rises. The law can be described using a demand schedule, a demand curve, and a demand function. A demand schedule is a table that describes the different quantities of a commodity demanded at different prices. A demand curve is the graphical representation of a demand schedule. Individual demand for a commodity is the amount a single consumer buys at a given price during a particular period of time. Market demand is the total quantity of a commodity that all households are willing to buy at a given price during a given period of time.

Supply is the quantity of a commodity that producers are willing to produce and offer for sale at a particular price during a given period of time. The law of supply states that, other things remaining the same, the quantity of a commodity that firms will produce and offer for sale rises with a rise in price and falls with a fall in price. A supply schedule is a tabular description that shows various quantities that producers are willing to produce and sell at various prices during a given period. Individual supply is the quantity of a commodity that one firm is willing to produce and offer for sale at a particular price during a given period of time. Market supply is the quantity of a commodity that all the producers are willing to produce and offer for sale at a particular price during a given period of time.

Market equilibrium is a situation in which the quantity demanded of a commodity equals the quantity supplied of the commodity. An equilibrium price is the price at which quantity demanded equals quantity supplied. An equilibrium quantity is the amount of a commodity bought and sold at an equilibrium price. Graphically, the equilibrium price is determined at the point of intersection of the demand curve and the supply curve.

Review Questions

Part I: Write 'True' if the statement is correct or 'False' if it is not correct for each of the following statements.

1. The only factor that can cause a change in the quantity demanded of a commodity is a change in the price of the commodity.
2. Demand schedule refers to a tabular representation of the relationship between price and quantity demanded.
3. A change in price of coffee will not affect the quantity demanded of coffee.
4. A change in price of wheat will affect the quantity stored of wheat.
5. Equilibrium price is the price at which quantity demanded equals quantity supplied.
6. An increase in quantity supplied of a commodity leads to a decrease in its price.
7. At equilibrium price, the quantity demanded and quantity supplied are not equal.
8. Market equilibrium is a situation in which quantity demanded of a commodity equals the quantity supplied of the commodity.
9. The amount of a commodity that is bought and sold at a lower price is called the equilibrium quantity.
10. Provided that market demand and supply remain unchanged, the price will neither tend to rise nor fall below this equilibrium price.

Part II: Choose the correct answer among the alternatives for the following questions.

1. Demand reflects the quantity of a commodity that consumers
   A . want to buy it at a different price,
   B . need to buy at alternative prices.
   C . are willing and able to buy at alternative prices.
   D . can be bought at alternative prices.
2. A demand schedule shows the relationship between the quantity demanded of a commodity over a given period of time and the ______________
   A . price of the commodity. C. income of the consumers.
   B . taste of the consumers. D. price of related commodities.
3. More of a commodity will be purchased at lower prices because
   A . consumers substitute this commodity for others whose price has not changed.
   B . at lower prices, consumers can buy more of this commodity with a given money income.
   C . more consumers will buy the commodity at lower prices than at higher prices.
   D . all of the above

4. A change in price of a commodity affects
   A . the quantity demanded of a commodity.
   B . consumers' willingness to buy the good.
   C . consumers' ability to buy the good.
   D . all of the above.
5. Which one of the following is not held constant in defining the demand schedule?
   A . Number of consumers C. Prices of other goods
   B . Price of the good in question D. Number of sellers in a market
6. Which of the following statements is not correct?
   A . Demand curves tell us how much consumers demand at different prices.
   B . Supply curves tell us how much producers supply at different prices.
   C . Demand and supply curves can tell us the actual price of a good in general.
   D . All of the above.
7. An economic system is said to be in equilibrium when.
   A . its important variables show no change
   B . no forces are acting on them to produce a change in their values
   C . A and B
   D . all of the above
8. The intersection of a market demand curve and a market supply curve for a commodity determines the
   A . equilibrium price for the commodity.
   B . equilibrium quantity for the commodity.
   C . point of neither increase nor decrease in price of the commodity.
   D . all of the above

Part III: Answer the following questions briefly and to the point.

1. Distinguish between the following pair of concepts:
   a. Individual demand and market demand
   b. Individual supply and market supply
2. What is the difference between demand and the quantity demanded for a commodity?
3. To get the market demand curve for a product, why do we add individual demand curves horizontally rather than vertically?
4. To get the market demand curve for a product, why do we add individual demand curves horizontally rather than vertically?
5. When is the market said to be in a state of equilibrium?